Integrative Enrichment of Protein Functional Annotations Using Large Language Models

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Abstract. Protein functional annotation relies on computational tools that differ in their methods, output formats and types of evidence produced. The results of these tools remain dispersed and therefore difficult to compare, inspect and reuse. This work proposes a pipeline that normalizes the outputs of six annotation tools, UPIMAPI, eggNOG-mapper, reCOGnizer, DeepFRI, DeepGO2 and CLEAN, into a canonical schema, enriches them with data from UniProt and InterPro and consolidates the evidence into an integrated per-protein profile. For each profile, a language model generates a structured summary that organizes the evidence by strength, flags conflicts and identifies gaps, guiding human review. The resulting profiles and summaries are made available through an interactive application that supports data visualization, navigation across proteins, filtering by metadata and evidence sources, detailed inspection of each profile and export of selected results. The integration pipeline was applied to a set of 1,802 proteins and the summarization stage was evaluated on a subset of 25 using two locally run models: Llama 3.1 (8B) and Qwen3.6-35B. Integration exposed corroboration between sources and tool-specific artifacts. In the summarization stage, Qwen3.6-35B identified EC number conflicts more consistently than Llama 3.1 (8B). Overall, the results suggest that multi-tool integration combined with large language model review can support the manual inspection of protein annotations using locally run models.

Keywords: Protein function · Data integration · UniProt · InterPro · Large language models · Protein profiles

1 Motivation

Protein functional annotation remains a major challenge in bioinformatics because many sequenced proteins still lack complete and reliable functional characterization. Although several computational tools can support this task, their outputs are usually heterogeneous, dispersed and difficult to compare directly. As a result, users must inspect several files manually to understand whether different tools agree, whether an annotation is supported by external database information or whether there are conflicts between sources. This work is motivated by the need to reduce that manual burden. It proposes a workflow that integrates previously generated annotation outputs, enriches them with structured information from

UniProt and InterPro and consolidates the available evidence into a single profile per protein. These profiles are then summarized with LLMs and made available through an interactive application, allowing users to inspect, compare and review the evidence more efficiently.

2 State of the Art

To understand the current context of bioinformatics it is important to consider the impact of the rapid growth of biological data and its implications for data analysis. The significant increase in data volume, combined with the diversity of available approaches and tools has contributed to the difficulty of obtaining consistent results, particularly in the field of protein functional annotation. This scenario highlights the need to understand the main existing approaches and their limitations.

2.1 Growth of Biological Data

Over the last years, bioinformatics has been increasingly influenced by the rapid growth of biological data. This growth is not only related to the amount of information being generated, but also to the complexity and heterogeneity of biological data [9]. The ability to acquire, process and analyse large biological datasets is now central to extracting useful knowledge from the available information [20]. Next-generation sequencing (NGS) has revolutionized genomics by enabling the parallel sequencing of millions of DNA fragments, with high processing capacity and improved cost-effectiveness [25]. It is characterized by the use of shorter reads and relies on massively parallel sequencing to obtain an accurate consensus sequence [28]. Advances in high-throughput DNA sequencing and bioinformatics have made it possible to systematically explore the genetic diversity present in complex microbial communities directly from environmental samples, without relying on prior isolation of individual microorganisms [27]. The growing amount of biological data has fuelled the development of omics approaches, transforming the way biology is studied and understood. These approaches analyse biological information across multiple layers at a large scale, enabling a more comprehensive understanding of organisms and their interactions [15]. Thus, NGS has facilitated the generation of complete genomic sequencing data, shifting the focus of research towards genomic annotation. When combined with metagenomics, it has significantly advanced research. However, the extraction of useful information from these data remains limited by the lack of functional information [11].

2.2 Protein Annotation Challenges

The exponential growth of biological data has resulted in a large number of protein sequences. Although the number of sequences has increased substantially, functional protein annotation has not kept pace with this growth. Approximately 50% of the proteins predicted across more than 500,000 sequenced genomes possess reliable functional annotations, leaving a major gap in the ability to interpret these data effectively [8]. Functional annotation of proteins is crucial for understanding biological mechanisms, as without accurate functional assignment, it is difficult to understand how proteins interact and the lack of rigorous annotation hinders the use of these sequences for practical applications, such as the development of new therapeutic treatments [13, 30]. Despite these limitations, computational methods for protein function prediction have advanced considerably. Traditional approaches, including sequence similarity-based methods, structural modeling and phylogenetic analysis, have long been used to infer protein function [21]. More recently, machine learning (ML) and deep learning (DL) techniques have been widely applied to predict functions based on existing data and have shown increasing effectiveness, particularly when integrated with experimental evidence [14].

2.3 Computational Tools for Protein Function Prediction

Computational methods have been developed across different methodological classes to support protein function prediction. Current approaches can be broadly grouped into three main categories: homology-based, structure-based and machine-learning-based methods. Although these approaches rely on different types of evidence, they all support the functional characterization of proteins that remain unannotated or only partially characterized. Recent studies have shown that protein function prediction benefits from integrating multiple types of evidence rather than relying on a single type alone [21, 22].

Within these categories, several open-source tools are available and generate different types of annotation outputs. Homology-based approaches typically produce results such as functional descriptions, conserved domain assignments, Gene Ontology (GO) terms, Enzyme Commission (EC) numbers, taxonomic information and pathway-related annotations. Representative examples of open-source tools in this group include reCOGnizer and UPIMAPI [26], as well as eggNOG-mapper [6]. Structure-based approaches provide a different type of information, often derived from predicted three-dimensional structures, structural comparisons, similarity searches or structure-informed functional inference. Tools such as Foldseek [16] and DeepFRI [12] are examples of methods that generate this type of output. Machine-learning-based approaches have also become increasingly relevant, particularly for proteins that are difficult to characterize using sequence similarity alone. Examples include DeepGO-based approaches [19, 17] as well as CLEAN [31].

Overall, these tools generate heterogeneous annotation outputs rather than a single unified type. This heterogeneity reflects the fact that protein function can be inferred from multiple biological perspectives, with each approach contributing a distinct layer of evidence. Consequently, integrating complementary evidence can improve the robustness and interpretability of functional predictions, but it also requires systematic organization and harmonization before results can be interpreted consistently.

2.4 Biological Databases for Protein Annotation

The integration of multiple information sources is essential for improving protein function prediction [18]. In this context, biological databases play a central role, despite the challenges associated with the massive growth of biological data and the need to make such data accessible to the scientific community [10]. However, the integration of large amounts of heterogeneous information from different databases and formats remains a major challenge [1]. One of the most relevant databases for protein annotation is UniProt, which aims to provide a comprehensive, high-quality and freely accessible collection of protein sequences annotated with functional information [3]. It integrates metadata from more than 180 interconnected databases and assigns stable protein identifiers [1].

Another resource of particular relevance is InterPro, which classifies protein sequences into families and predicts the presence of domains and functional sites by combining predictive models drawn from several established member databases, such as Pfam, PROSITE, SMART, CATH-Gene3D and PANTHER [4]. Each InterPro entry is assigned a stable accession and a set of manually curated Gene Ontology terms, which are propagated to every matched sequence [4, 5]. The complementarity of these databases is particularly relevant, as protein function prediction increasingly benefits from the integration of multiple types of evidence [21].

Given the large volume of results generated by annotation tools, manually consulting databases to add relevant information becomes inefficient. Database APIs enable automated queries and the retrieval of structured information in a standardized manner, facilitating the integration of large and heterogeneous bioinformatics datasets [7]. Both UniProt and InterPro provide REST APIs that allow programmatic access to their data, supporting identifier mapping, structured queries and the retrieval of functional annotations in reusable formats [1, 4].

2.5 Large Language Models for Data Harmonization

Large language models (LLMs) are trained on large volumes of text and are known for their ability to generate and process natural language [32]. Their relevance has grown in recent years, as they can support tasks such as information summarization, knowledge extraction and content organization [24]. LLMs are particularly important when information collected from different sources is expressed through distinct labels, descriptions or semantics and therefore needs to be aligned coherently. In this sense, they can help harmonize heterogeneous biological metadata by transforming inconsistent entries into more standardized representations [29]. In protein functional annotation, this capability becomes relevant because different tools may produce GO terms, EC numbers, functional descriptions, conserved domains or structural evidence that are not always presented uniformly. Although these results may be complementary, they can also be redundant, partially overlapping or semantically difficult to compare directly. After the initial results have been normalized and enriched with information retrieved from external databases, LLMs can be used to address this residual heterogeneity [23, 29]. In this way, LLMs appear here as a final semantic layer of harmonization. They support the integration and reorganization of the annotations already obtained, contributing to the production of a more consistent and interpretable final functional description for the proteins under study.

3 Objectives

This work aims to consolidate the dispersed outputs of several protein function analysis tools into a single evidence-aware profile per protein, enriched with structured and domain-level information from UniProt and InterPro. Its central goal is to use an LLM to transform each profile into a structured summary that highlights strong and weak evidence, conflicts and gaps for human review. Finally, the work aims to make the resulting profiles accessible through an interactive interface that supports navigation, visualization and export of the integrated evidence.

4 Materials and Methods

The work is built around a staged bioinformatics pipeline that uses the outputs of six functional annotation tools to generate an integrated profile for each protein. The pipeline starts from raw annotation files produced in a previous Snakemake-based study, where 1,802 proteins selected from the NCBI Protein database were processed with complementary functional annotation tools. The present work begins from those tool outputs and focuses on their normalization, enrichment, consolidation, LLM-based summarization and interactive exploration.

As shown in Figure 1, the pipeline is organized into three main stages, followed by an interactive application. The first stage normalizes the heterogeneous outputs of the six tools into a shared annotation format. The second stage enriches the normalized records with structured information from UniProt and InterPro, then consolidates all evidence into one profile per protein. The third stage submits selected consolidated profiles to LLMs, which generate structured summaries for human review. The final output is made available through an interactive application that allows users to filter, inspect, compare and export the resulting profiles. A summary of the data flow between stages is provided in Appendix F.

The pipeline follows three design principles. First, each piece of evidence is stored as an individual annotation record, so that the original source is preserved. Second, every record keeps the tool that produced it and the confidence level assigned during normalization. Third, LLMs are used only as a summarization layer. They organize the available evidence, highlight conflicts and missing information, but do not decide the final biological annotation.

4.1 Dataset and input tools

The working dataset consists of 1,802 protein sequences randomly selected from the NCBI Protein database in a previous study. In that study, the proteins

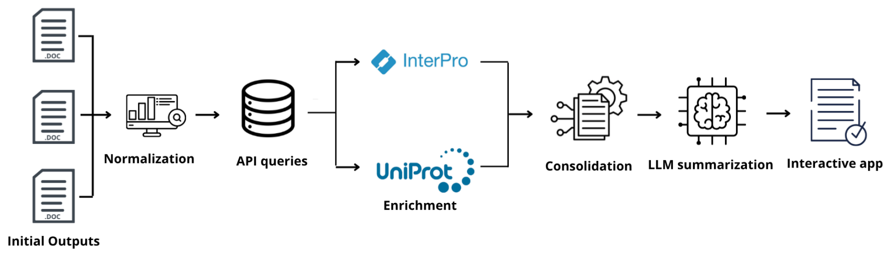

Fig. 1. Overview of the proposed pipeline. The outputs of six functional annotation tools are normalized, enriched with UniProt and InterPro information, consolidated into integrated protein profiles, summarized by LLMs and made available through an interactive application.

were annotated through an automated Snakemake pipeline using six functional annotation tools. The present work uses the raw output files produced by those tools as input. All subsequent processing, including normalization, enrichment, consolidation, LLM summarization and the interactive application, is part of this work.

The six tools cover complementary methodological classes that reflect the approaches described in the state of the art. UPIMAPI, eggNOG-mapper and reCOGnizer provide homology-based, orthology-based or domain-based evidence. DeepFRI, DeepGO2 and CLEAN provide machine-learning-based predictions. Table 1 summarizes the tools, their methodological class, output format and main annotation types.

Table 1. Summary of the six input annotation tools.

Tool	Class	Format	Main annotations
UPIMAPI	Homology	TSV	Names, descriptions, GO, EC
eggNOG-mapper	Orthology	TSV	GO, EC, KEGG, COG, descriptions
reCOGnizer	Domains	TSV/XLSX	Domains (Pfam, COG, KOG, SMART, TIGRFAM), EC
DeepFRI	ML (structure)	CSV/JSON	GO (MF)
DeepGO2	ML (sequence)	TSV	GO (BP, CC, MF)
CLEAN	ML (sequence)	CSV	EC

4.2 Normalization (Stage 1)

Normalization, the first stage of the pipeline, receives the raw output files of the six tools listed in Table 1. A dedicated parser for each tool reads its native format and converts every piece of evidence into a canonical record, named `AnnotationRecord`, with a fixed set of fields: the protein accession, the source tool, the annotation type (GO term, EC number, domain, description or pathway), the value and optional label, the original score and a categorical confidence level (high, medium or low).

The confidence level is assigned by applying tool-specific thresholds to the original scores, as shown in Table 2. For e-value-based tools, namely UPIMAPI, eggNOG-mapper and reCOGnizer, the thresholds follow commonly adopted cutoffs in sequence similarity and domain search practice [2]. For the ML-based tools, a score above 0.5 has been suggested as a significance threshold in the DeepFRI literature [12] and an analogous configurable rule was adopted for DeepGO2. For CLEAN, whose scoring is based on contrastive distance in an embedding space [31], a separate range was defined to reflect the different scale of its output. All thresholds are configurable and can be adjusted for specific datasets.

Table 2. Confidence levels by score type.

Tool	Score type	High	Medium	Low
UPIMAPI, eggNOG, reCOGnizer	e-value	$\leq$ 1e−50	1e−50 $< e \leq$ 1e−10	$>$ 1e−10
DeepFRI, DeepGO2	ML confidence (0 to 1)	≥ 0.7	$0.3 \leq s < 0.7$	< 0.3
CLEAN	ML confidence (0 to 1)	≥ 0.5	$0.1 \leq s < 0.5$	< 0.1

The normalization stage produces two JSON files. The first, `annotations.json`, stores the full set of canonical annotation records generated from the six input tools. The second, `proteins.json`, stores the protein accessions and basic metadata required to organize the following steps. Together, these files provide the structured input for the enrichment stage described next.

4.3 Enrichment (Stage 2)

The enrichment stage uses the normalized outputs produced in Stage 1 as its input. The two JSON files, `annotations.json` and `proteins.json`, provide the canonical annotation records and the list of protein accessions to be enriched. This stage adds structured information from UniProt and InterPro, then prepares the evidence for consolidation into a single profile per protein.

For UniProt, each protein accession is queried to retrieve available information such as review status, annotation score, protein and gene names, organism, functional description, GO terms, EC numbers, catalytic activity and sequence features. For InterPro, the pipeline retrieves family, domain and signature information associated with each protein, including entry type, source database, GO terms and sequence positions. A complete reference of the extracted fields is provided in Appendix A.

Enrichment was performed whenever the protein accession could be resolved in the external databases. When no UniProt or InterPro information was available, for example because an entry was inactive or could not be retrieved, the protein was kept in the dataset and the corresponding enrichment fields were left empty. This avoided removing proteins from the analysis only because external database information was incomplete.

The enrichment process also resolves GO terms that remained with an unknown aspect after normalization, using the aspect information available in the UniProt records. API access is controlled with rate limiting, checkpointing and local caching. This sub-step produces three files: the UniProt enrichment records, the InterPro enrichment records and the GO aspect map.

In the consolidation sub-step, the normalized records and enrichment records are merged by protein accession. Equivalent annotations reported by more than one source are grouped into a single entry while preserving all contributing sources and their scores. The resulting profile organizes the evidence into sections such as identity, function, GO annotations, enzymatic information, domains, features, sequence information, references and InterPro entries. It also stores an evidence summary with the tools present, the number of annotations, the confidence distribution and the overall confidence. The final output of this stage is `protein_profiles.json`, which contains one consolidated profile per protein.

4.4 LLM summarization (Stage 3)

This stage receives the `protein_profiles.json` file produced during consolidation. This file contains one consolidated profile per protein, including normalized annotations, enrichment information, confidence levels and provenance. Each selected profile is then submitted to an LLM, which generates a structured summary to support human review.

The LLM is not used to predict protein function directly. Instead, it is used to organize the evidence already present in the consolidated profile. The prompt instructs the model to use only the input data, preserve provenance, treat tool predictions as predictions and report conflicts or missing information when present. The complete prompt template is provided in Appendix B. An example of a consolidated profile used as input is shown in Appendix H.

The models were served locally through Ollama on a GPU server. Two locally run models were tested: Llama 3.1 (8B), used as a compact baseline and Qwen3.6-35B, used as the larger comparison model. Qwen3.6-35B was run in non-thinking mode with `Q4_K_M` quantization.

The output is constrained by a JSON Schema and validated after generation. The resulting summary contains seventeen fields covering identity, reported function, GO annotations, enzymatic activity, domains, pathways, tool predictions, strong and weak evidence, detected conflicts, missing information and review notes.

The manual evaluation was performed on a subset of 25 protein profiles selected from the consolidated dataset. These proteins were chosen to cover five representative profile types: reviewed proteins, poorly annotated proteins, EC prediction cases, conflicting evidence cases and ML-only or weak-evidence cases. The full list of test proteins is provided in Appendix G. Each profile was submitted independently to both models and the summaries were reviewed according to completeness, faithfulness to the input profile, clarity, evidence separation and caution. The evaluation rubric is provided in Appendix C.

4.5 Interactive application

The final component of the workflow is an interactive application developed in Streamlit. The application brings together the consolidated profiles produced in Stage 2 and the summaries generated in Stage 3, when available, matching both sources by protein accession. This allows the user to inspect the original evidence and the corresponding summary in the same environment.

The interface was designed to support manual exploration of the dataset rather than to replace biological review. Users can filter the proteins by accession, gene name, organism, review status, confidence level and evidence sources. These filters make it possible to focus on specific groups of proteins, such as poorly annotated entries, proteins with conflicting evidence or proteins with low overall confidence.

For each selected protein, the application provides access to the consolidated profile, the model-generated summary, GO annotations, domains, InterPro entries, sequence information, provenance and raw JSON data. This organization allows the reviewer to move from a global view of the dataset to the detailed evidence supporting a specific protein profile. The filtered subset can also be exported as a JSON file for later inspection or reuse. A detailed description of the interface layout is provided in Appendix D.

The implementation, including the normalization, enrichment, summarization scripts and the interactive application, is available at `https://github.com/luispedrosa94-dotcom/projeto-bioinformatica`.

5 Results

5.1 Integration results

The integration pipeline was applied to 1,802 proteins. Most entries were unreviewed in UniProtKB/TrEMBL (95.7%). The UniProt annotation score, which reflects the amount of functional information available for each UniProt entry, was generally low, with only 12 proteins reaching a score of 4 or above. After normalization, enrichment and consolidation, the pipeline produced one integrated profile per protein. Each profile contained an average of 129.9 annotations, with a median of 109 and a range from 21 to 682 annotations.

Integration made agreement between independent sources easier to inspect. Among the GO terms present in the consolidated profiles, 9,145 were supported by more than one tool, covering 1,410 proteins (78.2%). This shows that the pipeline can distinguish isolated predictions from annotations supported by multiple sources.

EC assignments were analysed separately because they provide a direct view of enzymatic function and often reveal disagreement between annotation sources. Figure 2 shows the distribution of distinct EC numbers per protein after consolidation.

The distribution shows that 967 proteins (53.7%) had more than one distinct EC number. This may reflect multifunctional enzymes, but it may also indicate disagreement between sources. In this dataset, 764 proteins (42.4%) had an EC number only from CLEAN, without support from other annotation sources. Only 280 proteins (15.5%) had EC support from reviewed UniProt information.

Overall per-protein confidence was mostly medium (41.4%) or low (40.5%), while only 18.1% of proteins were classified as high confidence. These results show that integration improves the organization and comparison of evidence,

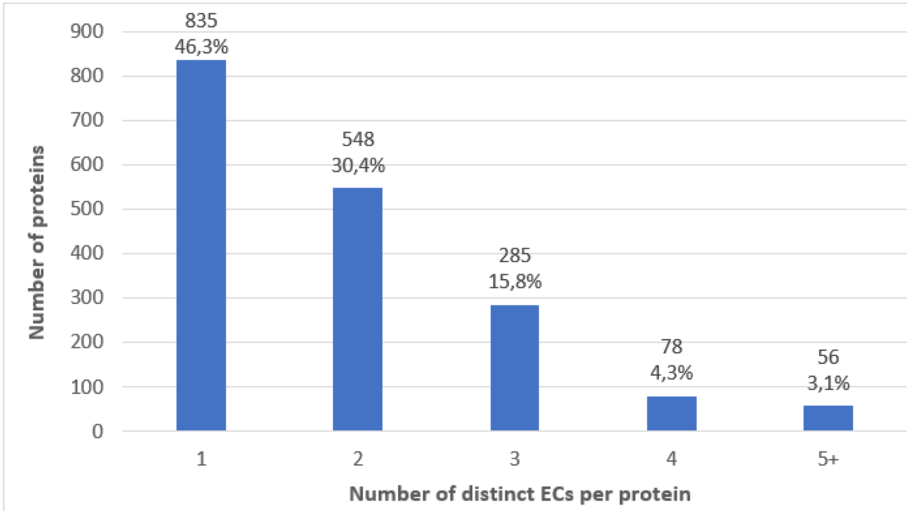

Fig. 2. Distribution of distinct EC numbers per protein in the consolidated dataset ($n = 1,802$). Each bar represents the number of proteins assigned a given count of distinct EC numbers after consolidation. Proteins with two or more distinct EC numbers correspond to 967 cases (53.7%) and were flagged for review in the summarization stage.

but does not remove the limitations of incomplete reviewed information or weak enrichment evidence. This motivates the LLM summarization step described in the next section. Detailed statistics by tool and annotation type are provided in Appendix E.

5.2 Evaluation of language model summarization

The LLM summarization stage was evaluated on 25 consolidated protein profiles selected from the full dataset. The subset was designed to include different review situations, such as reviewed entries, poorly annotated proteins, EC prediction cases, conflicting evidence cases and profiles mainly supported by weaker computational evidence. The purpose was not to infer new functions, but to assess whether the generated summaries could organize the available evidence in a clear and reliable way.

Both models produced structured outputs that followed the expected JSON format. Llama 3.1 (8B) generated shorter summaries, with generally readable output, but it sometimes omitted relevant uncertainty or conflicts between sources. Qwen3.6-35B produced more detailed summaries, with clearer separation between reviewed information, tool predictions and unsupported assignments. This difference was most visible in profiles with incomplete evidence or conflicting EC predictions. Table 3 presents representative examples from the 25-protein evaluation subset, highlighting how each model behaved in different evidence scenarios.

Table 3. Representative examples from the 25-protein evaluation subset.

Group (protein)	**Challenge**	**Llama 3.1 (8B)**	**Qwen3.6-35B**
Reviewed (Q46505)	Consistency with curation	Reports the curated EC, omits the divergent EC, conflict field empty	Identifies the conflict between the curated EC and the domain-based EC and interprets its biological cause
Poorly annotated (B3E8W6)	Absence of curated function	Notes the absence in two generic points	Notes the absence with five specific points, including no domains, no pathways and no localization
EC predictions (H1XYF8)	Multiple EC numbers from distinct sources	Lists five EC numbers without noting incompatibility	Flags that catalase and protein kinase assignments from eggNOG-mapper are biologically incompatible
Conflicting evidence (F6D5H3)	Contradictory EC and domains	Lists the EC numbers, conflict field empty	Flags EC and domain family conflicts, and notes that the function is inferred from the complex
ML-only / weak (J0HCT8)	Only predicted evidence	Lists the predictions, with generic notes	Detects the conflict between the transferase EC from CLEAN and the peptidase domain from reCOGnizer

Overall, the examples show that the summaries were most useful when they helped the reviewer navigate the consolidated evidence without replacing it. The strongest outputs preserved provenance, separated strong evidence from weaker predictions and explicitly reported missing or conflicting information. The main limitations appeared when the input profile contained little information, or when uncertain predictions were repeated with more confidence than warranted. A side-by-side comparison of model outputs for the same protein is provided in Appendix I.

To complement the evaluation, the interactive application was used to inspect proteins from the same 25-protein subset together with their summaries. Figure 3 shows the view for Q46505, a reviewed protein included in the evaluation subset. The interface combines the consolidated profile with the Stage 3 LLM summary, allowing the reviewer to check the generated text against the underlying evidence, including review status, annotation score, sequence length and overall confidence.

Taken together, these results indicate that LLM summarization can support the review of integrated protein profiles, especially when several evidence sources must be compared. The summaries help organize evidence and highlight uncertainty, but the final biological interpretation should remain with the human reviewer.

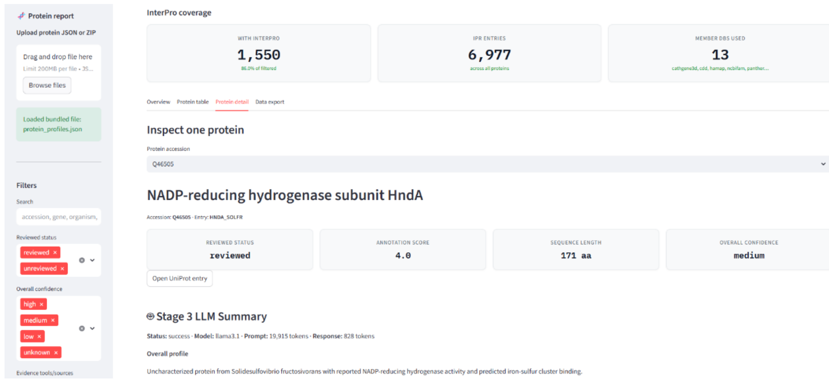

Fig. 3. Interactive application view for protein Q46505, included in the 25-protein evaluation subset. The interface combines protein-level metadata with the Stage 3 LLM summary, allowing the generated output to be inspected alongside the consolidated evidence available for that protein.

6 Conclusion

This work presented a workflow for integrating, enriching and reviewing protein functional annotations generated by complementary tools, including homology-based methods, structure-informed predictions and machine-learning-based approaches. The pipeline normalized heterogeneous outputs, enriched them with structured information from UniProt and InterPro and consolidated the available evidence into one profile per protein.

The results show that integration makes protein annotations easier to inspect by distinguishing corroborated evidence from isolated predictions and by exposing conflicts between sources. The enrichment stage added relevant biological context, including review status, annotation scores, functional descriptions, GO terms, EC numbers, domains and sequence features, although some proteins remained limited by incomplete external database information.

The LLM summarization stage helped organize complex profiles into structured summaries for human review. Qwen3.6-35B generally produced more complete and cautious summaries than Llama 3.1 (8B), especially in cases involving weak evidence or conflicting annotations. Still, the model should be seen as a review-support tool rather than an automatic annotator.

The interactive application connects the consolidated profiles and generated summaries in a single review environment, making the results easier to explore and verify. Overall, the workflow provides a practical framework for inspecting protein functional annotations while preserving provenance and transparency.

Future work should extend the summarization stage to all 1,802 protein profiles using both Llama 3.1 and Qwen3.6-35B, which was not possible here due to time and computational constraints. Further improvements should include a larger manual evaluation, refined confidence rules, additional biological validation sources and a cleaner interactive application with more flexible search fields.

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A Enrichment fields

This appendix summarizes the main fields retrieved during the enrichment stage from UniProt and InterPro.

Table 4. Fields retrieved from UniProt.

Field	Example
reviewed_status	unreviewed
annotation_score	1
protein_name	NADP-reducing hydrogenase subunit HndA
gene_name	hndA
organism	Solidesulfovibrio fructosivorans
protein_existence	Evidence at protein level
reported_function_description	Catalyzes the reduction of NADP...
keyword	Iron-sulfur
subcellular_location	Cytoplasm
GO_BP, GO_CC, GO_MF	GO:0006730, GO:0005737, GO:0016491

Table 5. Fields retrieved from InterPro.

Field	Additional metadata
interpro_entry	Type, source database, GO terms
interpro_unintegrated_signature	Type, source database, linked IPR entry

B Prompt template

The following prompt is sent to the language model for each protein. The placeholder `{protein_json}` is replaced with the complete consolidated profile of the protein being summarized.

```
/no_think
You are a protein profile summarization assistant.
Your task:
Given the complete JSON profile for ONE protein, write a structured
summary of the information contained in that profile.

Important framing:
- You are NOT predicting protein function from sequence or from your
  own knowledge.
- You are NOT choosing a final biological annotation.
- You are NOT acting as a curator who decides the correct function.
- You are summarizing what the record contains: identity, source
  status, reported functions, GO terms, EC numbers, domains, features,
  pathways, tool predictions, literature/curated evidence, weak
  evidence, missing information, and conflicts.
- The output should help a human quickly understand what is known,
  predicted, uncertain, or absent for this specific protein.

Rules:
1.  Use only information present in the input JSON.
2.  Do not invent functions, pathways, EC numbers, organisms,
    locations, domains, gene names, scores, or references.
3.  Do not produce a "recommended annotation" or claim that one
    function is the final correct answer.
4.  If the input contains a curated/reviewed function, describe it
    as "reported" or "curated in the record", not as your own
    conclusion.
5.  If the input contains EC numbers, GO terms, pathways, domains,
    or protein names, summarize them as information present in the
```

```
    record.
6.  Keep evidence provenance visible: distinguish curated/literature/
    database evidence from homology evidence, domain evidence,
    automatic annotations, ML predictions, and contextual evidence.
7.  Treat ML predictions from tools such as DeepGO2, DeepFRI, CLEAN,
    or similar tools as predictions, not confirmed facts.
8.  Treat enrichment, pathway, or contextual information as supporting
    context only, not direct proof of individual protein function.
9.  Mention sparse, missing, generic, low-confidence, or conflicting
    information explicitly.
10. Be concise but complete enough for a supervisor/biologist to
    inspect the protein quickly.
11. Return valid JSON only. Do not include Markdown, commentary, or
    text outside the JSON object.
12. For "conflicting_or_inconsistent_information", consider the
    following situations as conflicts that MUST be reported when
    present in the input:
    a. The same GO term annotated with different confidence levels
       by different tools.
    b. Different EC numbers assigned to the same protein by different
       tools.
    c. Tool predictions that contradict the curated UniProt function.
    d. Domain hits from different databases that imply incompatible
       biological functions.
    e. Mismatch between organism/taxonomy data and the predicted
       function.
13. For "review_notes", always include 1 to 4 short suggestions to
    help a human reviewer prioritize what to check.
14. When confidence levels are explicitly listed in the input, include
    them in the summary bullets in parentheses.
15. Final reminder: the goal is to help a human curator quickly
    understand: (a) what is known with confidence, (b) what is
    predicted but uncertain, (c) what is contradictory, and (d) what
    needs further checking.
16. "overall_profile_summary" MUST always be a single non-empty
    sentence (between 10 and 50 words).
```

C Evaluation rubric

The following rubric was used to guide the manual review of the language model summaries. Each criterion is scored from 0 (poor) to 3 (excellent).

Table 6. Summary evaluation rubric.

Criterion	0	1	2	3
Completeness	Major categories missing	Several important categories missing	Most categories covered, minor omissions	All relevant categories covered
Faithfulness	Clear unsupported claims or invented information	Some unsupported wording	Mostly faithful, minor wording issues	Fully faithful to the input profile
Clarity	Confusing or not useful	Somewhat readable but vague	Readable and mostly useful	Clear, concise, and useful for a biologist
Evidence separation	Does not separate evidence types	Weak separation, mixes predictions with curated	Mostly separates evidence types	Clearly separates curated, predicted, weak, and conflicting
Caution	Overclaims or presents predictions as facts	Some overconfident wording	Mostly cautious	Consistently cautious, notes limitations

D Interactive application layout

The Streamlit application organizes the information across a sidebar and four main tabs.

The sidebar filters include accession, gene name, organism, GO term, domain, review status, overall confidence, tools, and evidence sources.

Table 7. Application tabs and their content.

Tab	Content
Overview	Summary metrics and distribution charts, including organism, confidence, tools, evidence sources, InterPro entries, tool predictions, and detected conflicts. These charts are generated with Plotly.
Table	Filterable list of proteins with expandable per-protein detail.
Detail	Single-protein view with sub-tabs for model summary, GO annotations, domains, InterPro, sequence, references, provenance, and raw JSON. The view is accompanied by quality-control metrics and validation warnings.
Export	Download of the filtered subset as a JSON file.

E Descriptive statistics of the consolidated dataset

This appendix provides additional descriptive statistics for the consolidated dataset, including annotation records by tool, records by annotation type, enrichment records, and confidence distribution.

Table 8. Annotation records by tool.

Tool	Records	%	Note
DeepGO2	131,072	56.0%	87% of all low-confidence records
eggNOG-mapper	54,833	23.4%	58% of all high-confidence records
reCOGnizer	26,662	11.4%	
UPIMAPI	10,561	4.5%	
DeepFRI	8,660	3.7%	
CLEAN	2,243	1.0%	
Total	**234,031**	**100%**	

Table 9. Records by annotation type.

Type	Records
GO terms	169,623
Protein descriptions	12,366
Pfam	8,502
EC numbers	5,245
COG	2,218
KOG	1,913

Table 10. Enrichment records.

Source	Records	Detail
UniProt	19,403	
InterPro	19,919	6,977 IPR entries and 12,942 member-database signatures

Table 11. Overall confidence distribution.

Level	By record	By protein
High	37.3%	326 proteins (18.1%)
Medium	20.9%	746 proteins (41.4%)
Low	41.7%	730 proteins (40.5%)

F Data flow between pipeline stages

This appendix summarizes the input, process, and output of each pipeline stage.

Table 12. Input, process, and output of each pipeline stage.

Stage	Input	Process	Output
Normalization (Stage 1)	Raw output files of the six tools	Parses each tool into the canonical schema, normalizes identifiers, and assigns categorical confidence	`annotations.json`, `proteins.json`
Enrichment (Stage 2)	`annotations.json`, `proteins.json`	Queries the UniProt and InterPro APIs and resolves GO aspects	`uniprot_enrichment.json`, `interpro_enrichment.json`, `go_aspect_map.json`
Consolidation (Stage 2)	Stage 1 annotations and enrichment files	Merges records by accession into a nested per-protein profile, groups equivalent items, and computes the evidence summary	`protein_profiles.json`
LLM summarization (Stage 3)	`protein_profiles.json` for the 25-protein subset	Inserts each profile into the prompt, generates a JSON Schema-constrained summary, and validates the output	`stage3_results.jsonl`
Interactive application	`protein_profiles.json` and `stage3_results.jsonl`	Joins profile and summary by accession	Streamlit interface

G Test protein subset

The 25 proteins used to evaluate the summarization stage were selected to cover five distinct profile types. Table 13 lists each protein with its accession, group, and organism.

Table 13. Test protein subset.

Accession	Group	Organism
Q46505	Reviewed	Solidesulfovibrio fructosivorans
Q2LXU2	Reviewed	Syntrophus aciditrophicus
Q2RI41	Reviewed	Moorella thermoacetica
P11560	Reviewed	Methanothermobacter marburgensis
P06131	Reviewed	Methanobacterium formicicum
B3E8W6	Poorly annotated	Trichlorobacter lovleyi
I9KF72	Poorly annotated	inactive UniProt entry
G8TMW4	Poorly annotated	Niastella koreensis
Q01Q37	Poorly annotated	Solibacter usitatus
A1ATR7	Poorly annotated	Pelobacter propionicus
U6EEC5	EC predictions	not available in record
G3F6H2	EC predictions	not available in record
H1YXH1	EC predictions	Methanoplanus limicola
A8UKK6	EC predictions	not available in record
H2EIF6	EC predictions	Malus domestica
H1XYF8	Conflicting evidence	Caldithrix abyssi
F6D5H3	Conflicting evidence	Methanobacterium paludis
U6EF42	Conflicting evidence	not available in record
K2R6I7	Conflicting evidence	Methanobacterium formicicum
F0T7S9	Conflicting evidence	Methanobacterium lacus
U6EF83	ML-only / weak evidence	inactive UniProt entry
H0UK06	ML-only / weak evidence	Jonquetella anthropi
W2LNG5	ML-only / weak evidence	Phytophthora nicotianae
L0DDH1	ML-only / weak evidence	Singulisphaera acidiphila
J0HCT8	ML-only / weak evidence	Rhizobium leguminosarum

H Example consolidated profile

The following is a truncated view of the consolidated profile for protein B3E8W6, a poorly annotated protein from *Trichlorobacter lovleyi*. This example illustrates the structure provided to the language model. The full profile contains additional GO terms, features, and evidence details.


```
{
  "accession": "B3E8W6",
  "identity": {
    "protein_name": "Uncharacterized protein",
    "gene_name": null,
    "reviewed_status": "unreviewed",
    "annotation_score": 1.0,
    "protein_existence": "4: Predicted",
    "organism": {
      "scientific_name": "Trichlorobacter lovleyi
      (strain ATCC BAA-1151 / DSM 17278 / SZ)"
    }
  },
  "function": {
    "description": [],
    "keywords": ["Reference proteome", "Signal"],
    "subcellular_location": [],
    "catalytic_activity": []
  },
  "go_annotations": {
    "molecular_function": [
      {
        "go_id": "GO:0005488",
        "sources": ["deepgo2"],
        "source_details": {
```

```
        "deepgo2": {
          "source_type": "ml_sequence",
          "confidence": "medium",
          "score": 0.607
        }
      },
      "confidence": "medium"
    }
  ]
},
"enzymatic": {
  "ec_numbers": [
    {
      "ec_id": "4.2.2.17",
      "sources": ["clean"],
      "confidence": "low"
    }
  ]
}
}
```


This profile has no curated function, no gene name, and an annotation score of 1. The only enzymatic annotation shown in the excerpt comes from CLEAN with low confidence, while the GO term shown is an ML prediction from DeepGO2.

I Example LLM outputs

The following tables show selected fields from the structured summaries produced by each model for the same protein, B3E8W6. This protein was selected because it has no curated function and contains mostly predicted or limited evidence.

Table 14. Selected fields from the Llama 3.1 (8B) output for B3E8W6.

Field	Content
Overall summary	Uncharacterized protein from *Trichlorobacter lovleyi* with predicted molecular functions and cellular components by DeepGO2 tool.
Reported function	No curated function reported in the record.
Enzymatic summary	EC 4.2.2.17 predicted by CLEAN tool.
Conflicts	Empty.
Missing or limited information	No curated function reported. No EC numbers reported in the record.
Review notes	Verify the predicted enzyme activity against literature references. Check if the signal and chain features cover the full sequence or only a partial region.

Table 15. Selected fields from the Qwen3.6-35B output for B3E8W6.

Field	Content
Overall summary	Uncharacterized protein from *Trichlorobacter lovleyi* with no curated function, featuring a predicted signal peptide and multiple low-confidence ML predictions for molecular function and cellular location.
Reported function	No curated function, catalytic activity, or biological role is reported in the record. The protein is explicitly labeled as "Uncharacterized protein". Keywords include "Reference proteome" and "Signal".
Enzymatic summary	EC 4.2.2.17 predicted by CLEAN with low confidence.
Conflicts	Empty.
Missing or limited information	No curated function, EC numbers, or GO terms from literature or databases. No domain architecture from Pfam or InterPro identified. No subcellular location curated. No pathway assignments.
Review notes	Verify if the ML-predicted EC 4.2.2.17 is plausible given the absence of supporting domains or curated evidence. Check if the high-confidence DeepGO2 prediction for GO:0110165 aligns with the signal peptide prediction. Investigate why no Pfam or InterPro domains were detected.

In this case, both models identified the absence of curated function, but Llama 3.1 (8B) produced an internal inconsistency by listing the CLEAN EC prediction while also stating that no EC numbers were reported. Qwen3.6-35B produced a more specific and internally consistent summary, listing concrete types of missing information and review notes grounded in the profile.